Week 8 Homework

Exercise

What is Reve's Puzzle? Describe it, and describe the Frame-Stewart Algorithm in general and using an example.

Exercise

Use generating functions to solve:
a. $a_n = 6a_{n-1} - 8a_{n-2} + 3$ with $a_0 = 1$ and $a_1 = 0$
b. $a_n = 3a_{n-1} + 4^{n-1}$ and $a_0 = 1$

Exercise

Use generating functions to solve:
a. $a_n = \frac{1}{2}a_{n-1} + 1$ with $a_1 = 1$
b. $b_n = 4b_{n-1} - 3b_{n-2} + 2^n$ with $b_1 = 1$ and $b_2 = 11$

Exercise

Find the closed form solution to $a_n = a_{n-1} + 2(n-1)$ with $a_1 = 2$.

Exercise

Solve the following recurrence relations:
Find a recurrence relation for the number, t_n of length n ternary strings with no two consecutive 1s. What are the initial conditions? What is the generating function for t_n?

Exercise

Solve the following recurrence relations:
Find a recurrence relation for the number, d_n of length n ternary strings with no two consecutive 1s or no two consecutive 2s. What are the initial conditions? What is the generating function for d_n? (Though very similar to the previous problem, this version is much harder. Hint: The solution is not $d_n = d_{n-1} + 2d_{n-2}$, that is only part of the solution.)

Exercise

Consider tiling a $2 \times n$ board with 2×1 and 2×2 tiles.
a. Let d_n count the number of tilings of where the 2×1 can only be positioned vertically and 2×1 cannot be adjacent. Find a recursive relationship for d_n and find initial conditions for d_n.
b. Let g_n count the number of tilings of where the 2×1 can only be positioned vertically and 2×2 cannot be adjacent. Find a recursive relationship for g_n and find initial conditions for g_n.

Exercise

Let g_n count the number of subsets of $\{1, 2, 3, \ldots, n\}$ containing no two consecutive integers. For example, $g_2 = 3$, the empty set, $\{1\}$, and $\{2\}$. Find a recurrence relationship and generating function for g_n.

Exercise

Solve the recurrence relation $a_n = 2a_{n-1} - a_{n-2}$. Find the solution when $a_0 = 1$ and $a_1 = 2$.
What do the initial terms need to be in order for $a_9 = 30$?
For which x are there initial terms which make $a_9 = x$?

Exercise

Consider the recurrence relation $a_n = 4a_{n-1} - 4a_{n-2}$. Find the general solution to the recurrence relation (beware the repeated root).
Find the solution when $a_0 = 1$ and $a_1 = 2$.
Find the solution when $a_0 = 1$ and $a_1 = 8$.

Exercise

Define the Lucas numbers as $L_n = L_{n-1} + L_{n-2}$ with $L_1 = 1$ and $L_2 = 3$. What is the generating function for this sequence? Find a closed from solution for L_n.

Exercise

Let $f_n = f_{n-1} + f_{n-2}$ with $f_0 = 0$ and $f_1 = 1$. Define $F_n = F_{n-1} + F_{n-2} + f_n$ for $n \geq 1$. Express F_n in terms f_n and f_{n+1}.

Proof

Consider the sequence $f_n = f_{n-1} + f_{n-2}$ with $f_0 = 0$ and $f_1 = 1$. Prove $f_0 + f_1 + f_2 + f_3 + \cdots f_n = f_{n+2} - 1$.

Proof

Consider the sequence $d_n = (n-1)(d_{n-1} + d_{n-2})$ with $d_1 = 0$ and $d_2 = 1$. Prove that $d_n > (n-1)!$ for $n \geq 4$.

Proof

Prove $\sum_{k=0}^{n} \binom{n}{k} 4^k = 5^n$.

Proof

Suppose that a particular real number x has the property that $x + \frac{1}{x}$ is an integer. Prove that $x^n + \frac{1}{x^n}$is an integer for all natural numbers n.

Proof

Let T_n be the triangular numbers defined by $T_n = T_{n-1} + n$ with $T_0 = 0$ and n, p, and q be positive integers. Prove that if $T_n = pq$ then $T_{n+p+q} = T_{n+p} + T_{n+q}$.

Proof

Let T_n be the triangular numbers defined by $T_n = T_{n-1} + n$ with $T_0 = 0$ and n, p, and q be positive integers. Prove that if $T_{n+p+q} = T_{n+p} + T_{n+q}$ then $T_n = pq$.

Proof

Consider the sequence $f_n = f_{n-1} + f_{n-2}$ with $f_1 = 1$ and $f_2 = 2$. Calculate $f_1 + f_3 + f_5 + f_7 + \cdots + f_{2n-1}$ for small n to create a hypothesis for the sum of the first n odd indexed Fibonacci numbers, then prove this formula.

Proof

Consider the sequence $f_n = f_{n-1} + f_{n-2}$ with $f_1 = 1$ and $f_2 = 2$. Find (like the preceding problem) and prove formulas for:
$f_0 + f_2 + f_4 + f_6 + \cdots f_{2n}$

Proof

Consider the sequence $f_n = f_{n-1} + f_{n-2}$ with $f_1 = 1$ and $f_2 = 2$. Find (like the preceding problem) and prove formulas for:
$f_1 - f_2 + f_3 - f_4 + \cdots (-1)^{n-1} f_n$

Proof

Consider the sequence $d_n = 2d_{n-1} + d_{n-2}$ with $d_0 = 1$ and $d_1 = 3$. Prove

$$d_n = \sum_{i \geq 0} 2^i \binom{n+1}{2i}$$

is a solution to this recurrence relationship.

Proof

Using the definition for $\binom{n}{k} = \frac{n(n-1)(n-2)\cdots(n-k+1)}{k!}$ when $n < 0$, prove

$$\binom{-1/2}{n} = \frac{\binom{2n}{n}}{(-4)^n}$$

Proof

Let F_n be the Fibonacci sequence with $F_1 = 1$ and $F_2 = 1$. Prove $F_n \leq (\frac{5}{3})^n$ for all positive integers.

Proof

The indicated diagonals of Pascal's triangle sum to Fibonacci numbers. Prove this patterns continues forever.
Fibonacci and Pascal